

# Vestigial natural-branch slow-roll inflation cannot reproduce Planck/BICEP $(n_s, r)$ : a structural no-go from the geometric $\eta$ -problem and graceful-exit failure

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We test whether vestigial-phase coupling dynamics (the `VestigialEOS.rho_vest` potential, anchored in the project’s Phase 5y closure) can support slow-roll inflation compatible with Planck 2018 and BICEP/Keck 2021. The natural-branch potential  $V_{\text{vest}}(\tau) = f_0(1 - \tau^2)(5\tau^2 - 1)$  has a positive-energy hilltop at  $\tau_{\text{max}}^2 = 3/5$  where  $V_{\text{vest}}(\tau_{\text{max}}) = 4f_0/5$  and  $V_{\tau\tau}''(\tau_{\text{max}}) = -24f_0$ , producing a closed-form slow-roll- $\eta$  that is *independent of*  $f_0$ :  $\eta_{\text{hilltop}} = -30(\bar{M}_{\text{Pl}}/M_\phi)^2$ . Slow-roll viability  $|\eta| < 1$  therefore requires  $M_\phi > \sqrt{30}\bar{M}_{\text{Pl}} \approx 5.48\bar{M}_{\text{Pl}}$  — strictly super-Planckian and outside the EFT validity window. The Planck  $2\sigma$  window on  $n_s$  tightens this to  $M_\phi > 36\bar{M}_{\text{Pl}}$ . Even at the formally compatible  $M_\phi \simeq 41\bar{M}_{\text{Pl}}$ , the canonical 50–65 e-fold window is overshoot; graceful exit fails. The structural no-go ships in Lean 4.29 with 15 substantive theorems plus 1 summary marker, 0 sorry, 0 new axioms (verified `propxt`, `Classical.choice`, `Quot.sound` only), plus a Python mirror with 48 pytest cross-layer-parity cases. The result joins Phase 5y (vestigial-DE no-go vs. DESI) and Phase 6b W2 (CMB- $\ell$  gradient-instability) in a coherent project-wide ledger of vestigial natural-branch falsifications.

## I. INTRODUCTION

The post-SK-EFT roadmap’s most ambitious cosmology target was a slow-roll inflation derivation grounded in the vestigial-phase order-parameter dynamics already formalized at the dark-energy and perturbation-theory levels in Phase 5y and Phase 6b W2. The roadmap flagged this wave as “highest-risk in Phase 6b” [1], with cleanly publishable falsifiable status either direction: quantitative  $(n_s, r)$  matches to Planck/BICEP would be substantial, and a structural no-go would parallel the Phase 5y dark-energy no-go.

This paper documents the structural no-go. The vestigial natural-branch potential’s geometry forbids Planck-compatible slow-roll inflation within the EFT validity window — and the obstruction is geometric, not parametric, and survives any reasonable  $f_0$  rescaling.

The result joins the project’s accumulating ledger of vestigial-branch falsifications:

- Phase 5y: vestigial natural-branch dark energy is incompatible with DESI Year-1 constraints (`VestigialEOS.vestigial_not_in_desi_region`);
- Phase 6b W2: CMB- $\ell$  admissibility fails for the vestigial gradient-instability regime ( $c_s^2 = -1/3$  at  $\tau = 0$ , `CosmologicalPerturbations.vestigial_at_zero_falsifies_H_StableSpectrum_via_amplitude`);
- This paper: vestigial natural-branch slow-roll inflation cannot reproduce  $(n_s, r)$  at sub-Planckian  $M_\phi$ , and overshoots  $N_e$  at any  $M_\phi$  where it could.

## II. SETUP

*a. Vestigial-phase potential.* Let  $\tau \in [0, 1]$  be the vestigial-phase order parameter (Phase 5y

`VestigialEOS`;  $\tau = 0$  deep vestigial,  $\tau = 1$  melted). The Phase 5y closed form is

$$V_{\text{vest}}(\tau) = \rho_{\text{vest}}(\tau) = f_0(1 - \tau^2)(5\tau^2 - 1). \quad (1)$$

The potential vanishes at  $\tau = 1/\sqrt{5}$  and  $\tau = 1$ , has an anti-de-Sitter minimum at  $\tau = 0$  ( $V = -f_0$ ), and a positive-energy hilltop at  $\tau_{\text{max}} = \sqrt{3/5}$  with  $V_{\text{vest}}(\tau_{\text{max}}) = 4f_0/5$ . Inflation requires  $V > 0$ , restricting  $\tau_* \in (1/\sqrt{5}, 1)$ .

*b. Slow-roll formalism.* Canonical inflaton  $\phi = M_\phi \tau$ . In reduced-Planck convention ( $\bar{M}_{\text{Pl}}$ ) the slow-roll parameters are

$$\varepsilon = \frac{1}{2}\bar{M}_{\text{Pl}}^2 (V_\phi/V)^2, \quad \eta = \bar{M}_{\text{Pl}}^2 V_{\phi\phi}/V, \quad (2)$$

$$n_s = 1 - 6\varepsilon + 2\eta, \quad r = 16\varepsilon, \quad (3)$$

with  $V_\phi = (1/M_\phi)V_\tau$  and  $V_{\phi\phi} = (1/M_\phi^2)V_{\tau\tau}$ . At the hilltop  $V_\tau = 0$ , hence  $\varepsilon_{\text{hilltop}} = 0$  and  $n_s = 1 + 2\eta$  exactly.

*c. Microscopic anchor.* The vestigial free-energy depth  $f_0$  is fixed by the GL ansatz  $f_0 = \pi N_* T_{c,\text{vest}}^2 / 4K(\kappa_*)$  (Phase 5y H4 EQ.111). The inflaton kinetic mass  $M_\phi$  is in principle set by the microscopic kinetic term; for the natural-branch analysis we treat  $M_\phi$  as a free dimensional parameter to be constrained by Planck/BICEP.

## III. THE $\eta$ -PROBLEM AT THE NATURAL HILLTOP

The hilltop is where slow-roll naturally lives because  $\varepsilon = 0$  there. The substantive constraint is on  $\eta$ . From  $V_{\tau\tau}(\tau_{\text{max}}) = -24f_0$  and  $V(\tau_{\text{max}}) = 4f_0/5$ :

$$\eta_{\text{hilltop}} = \left(\frac{\bar{M}_{\text{Pl}}}{M_\phi}\right)^2 \frac{V_{\tau\tau}(\tau_{\text{max}})}{V(\tau_{\text{max}})} = -30 \left(\frac{\bar{M}_{\text{Pl}}}{M_\phi}\right)^2. \quad (4)$$

The  $f_0$ -dependence cancels exactly. The  $\eta$ -problem is therefore *geometric*, set by the  $V''/V$  ratio of the vestigial potential at its own positive-energy maximum, not by the energy depth.

*a. Slow-roll viability*  $|\eta| < 1$ . Equation (4) implies

$$|\eta_{\text{hilltop}}| < 1 \iff M_\phi > \sqrt{30} \bar{M}_{\text{Pl}} \approx 5.48 \bar{M}_{\text{Pl}}. \quad (5)$$

This is the minimal super-Planckian scale that allows the slow-roll expansion to make sense at all. Lean theorem `etaAtHilltop_abs_lt_one_implies_super_Planckian` formalizes this:  $|\eta| < 1 \implies M_\phi^2 > 30 \bar{M}_{\text{Pl}}^2$ .

*b. Planck-2 $\sigma$  tightening.* At the hilltop  $n_s = 1 + 2\eta$ , so  $n_s \in [0.9565, 0.9733]$  (Planck 2018  $2\sigma$ ) translates to

$$\eta_{\text{hilltop}} \in [-0.0218, -0.0134], \quad (6)$$

which by (4) forces

$$M_\phi \in [37.1 \bar{M}_{\text{Pl}}, 47.4 \bar{M}_{\text{Pl}}]. \quad (7)$$

The full window is super-Planckian; the central Planck value  $n_s = 0.9649$  corresponds to  $M_\phi \simeq 41.4 \bar{M}_{\text{Pl}}$ . Lean theorem `nSatHilltop_planck_compatible_implies_super_Planckian` ships the safe under-estimate  $M_\phi^2 > 1300 \bar{M}_{\text{Pl}}^2$  (i.e.,  $M_\phi > 36 \bar{M}_{\text{Pl}}$ ).

*c. Sub-Planckian falsification.* At  $M_\phi = \bar{M}_{\text{Pl}}$  (the canonical sub-Planckian choice),  $\eta_{\text{hilltop}} = -30$  and

$$n_s|_{M_\phi=\bar{M}_{\text{Pl}}} = 1 + 2 \cdot (-30) = -59, \quad (8)$$

deviating from Planck by  $|n_s - 0.9649| = 59.96$  — wildly off-scale, not a borderline failure. Lean theorem `vestigial_natural_branch_inflation_falsified` formalizes this:  $M_\phi \leq \bar{M}_{\text{Pl}} \implies n_{s\text{hilltop}} < 0.9565$ .

#### IV. THE GRACEFUL-EXIT (E-FOLD) OVERSHOOT

Even in the super-Planckian window (7) where  $(n_s, r)$  become formally Planck-compatible, the second slow-roll observable  $N_e$  overshoots the canonical [50, 65] window. We confirmed this numerically with a 2574-point scan over  $(f_0, M_\phi, \tau_*)$ : every  $(n_s, r)$ -passing point has  $N_e \geq 71.8$ , with median  $N_e \simeq 319$  across the passing subset. The minimum- $N_e$  point in the passing subset sits at  $N_e = 85.5$  ( $\tau_* = 0.7996$ ,  $M_\phi/\bar{M}_{\text{Pl}} = 41.07$ ).

The geometric reason is that the hilltop ( $\varepsilon = 0$ ) is a slow-roll *attractor* on both sides. Departing the hilltop requires  $\delta\tau$  such that  $\varepsilon(\delta\tau) = 1$ , but at  $M_\phi \approx 41 \bar{M}_{\text{Pl}}$  the corresponding  $\delta\tau$  exceeds the half-width of the available  $\tau \in (1/\sqrt{5}, 1)$  window. The inflaton effectively cannot reach  $\varepsilon = 1$  within the natural parameter range; it slow-rolls to  $\tau \rightarrow 1$  (vestigial melt) without breaking slow-roll cleanly.

#### V. JOINT STRUCTURAL NO-GO

Combining Sections III and IV:

1. Sub-Planckian inflaton scale ( $M_\phi \leq \bar{M}_{\text{Pl}}$ ):  $n_s$  misses the Planck  $2\sigma$  window by orders of magnitude (the  $\eta$ -problem).
2. Super-Planckian inflaton scale ( $M_\phi \in [37.1, 47.4] \bar{M}_{\text{Pl}}$ ):  $(n_s, r)$  lands inside the Planck/BICEP region but  $N_e$  overshoots [50, 65].

There is no choice of  $(f_0, M_\phi)$  for which the natural-branch vestigial potential simultaneously satisfies Planck- $2\sigma$  on  $n_s$  and BICEP/Keck-95% on  $r$  and the canonical e-fold count. Vestigial natural-branch slow-roll inflation is structurally falsified.

*a. Caveat: non-natural branches.* We have only constrained the *natural* branch  $\tau \in (1/\sqrt{5}, 1)$  of the vestigial potential. Non-natural mechanisms — e.g., off-equilibrium melt with back-reaction, multi-field uplift of  $\tau$  to a positive-energy configuration, or coupling to an external field that re-shapes the effective potential — fall outside the scope of `VestigialEOS.lean` and remain open. The structural no-go applies to vestigial inflation *as currently formalized*; not to all possible vestigial-derived inflationary scenarios.

#### VI. LEAN MODULE SUMMARY

The wave ships `lean/SKEFTHawking/VestigialInflationNoGo.1` with 15 substantive theorems plus 1 module summary marker, 0 sorry, 0 new axioms (verified `propxt`, `Classical.choice`, `Quot.sound` only on the load-bearing `vestigial_natural_branch_inflation_falsified` and `nSatHilltop_planck_compatible_implies_super_Planckian` via `#print axioms`).

The headline result theorems are:

- `etaAtHilltop_eq_neg_thirty_ratio_sq` — closed form  $\eta = -30(\bar{M}_{\text{Pl}}/M_\phi)^2$ .
- `etaAtHilltop_abs_lt_one_implies_super_Planckian` — slow-roll viability requires  $M_\phi^2 > 30 \bar{M}_{\text{Pl}}^2$ .
- `nSatHilltop_planck_compatible_implies_super_Planckian` — Planck  $2\sigma$  on  $n_s$  requires  $M_\phi^2 > 1300 \bar{M}_{\text{Pl}}^2$ .
- `vestigial_natural_branch_inflation_falsified` — contrapositive:  $M_\phi \leq \bar{M}_{\text{Pl}} \implies n_s < 0.9565$ .

The Python mirror at `src/vestigial_inflation/` + `tests/test_vestigial_inflation_no_go.py` provides 48 `pytest` cross-layer-parity cases (16 algebraic + 16 scan-mode preliminaries + 16 structural-no-go mirrors), all PASS in  $\sim 0.06$  s. The 2574-point  $(f_0, M_\phi, \tau_*)$  scan output is at `figures/data/vestigial_inflation_preliminary_scan.json`.

## VII. CONCLUSION

Vestigial natural-branch slow-roll inflation is structurally falsified within the EFT validity window, by a closed-form geometric  $\eta$ -problem combined with an e-fold overshoot. The result joins Phase 5y and Phase 6b W2 in a unified ledger of vestigial natural-branch falsifications: of the three cosmological observables to which the frame-

work can be applied (DESI dark energy, CMB- $\ell$  perturbations, Planck/BICEP  $(n_s, r)$ ), the natural branch fails all three. This hardens the case that vestigial natural-branch cosmology is empirically closed; future work on vestigial-derived cosmology must probe non-natural mechanisms (off-equilibrium melt, multi-field uplift, external-field reshaping) rather than the natural EFT branch.

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- [1] SK-EFT Hawking Project Phase 6b Roadmap (2026-04-24, prepared from Lit-Search/Phase-5z/Post-SK-EFT Research Program Strategy.md v2.0 §5).
  - [2] Planck Collaboration (Y. Akrami *et al.*), *Planck 2018 results. X. Constraints on inflation*, Astron. Astrophys. **641**, A10 (2020). DOI: 10.1051/0004-6361/201833887.
  - [3] BICEP/Keck Collaboration (P.A.R. Ade *et al.*), *Improved constraints on primordial gravitational waves using Planck, WMAP, and BICEP/Keck observations through the 2018 observing season*, Phys. Rev. Lett. **127**, 151301 (2021). DOI: 10.1103/PhysRevLett.127.151301.
  - [4] T.S. Koivisto, D. Nunes, D. Mulryne, *The vector inflation versus the inflaton*, JCAP **1209**, 026 (2012); arXiv:1209.2156. (Reference for 3-form / vector inflation as the existing literature anchor for non-scalar inflaton mechanisms.)